

Order guide

A Major category
TJ Category
X Series
L45 1 Maximum Compression
P 2 Cost Performance
D2 3 Outside Diameter
L5 4 Length

Please refer to the [series code](#) to choose your [models](#) and [parameters](#).

Code	Datum Hole	Place of Origin	Cost ^② Performance	Material	Maximum Compression ^①	Spring Constant Tolerance
ATJX-L45-P	Outer Diameter	Domestic	Standard	SUS304-WPB	Allowable Displacement L×45%	±10%
ATJX-L5-P	Reference SS				Allowable Displacement L×(50~60)%	

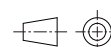
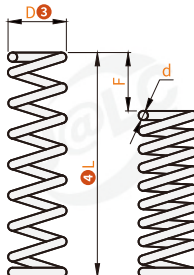


Dimensional Tolerance

D	Tolerance
Below 10	0 -0.5
Above 12	0 -0.8

Dimensional Tolerance

L	Tolerance
Below 50	± 1.5
Above 55	± 2.5



Perspective standard:
firstperspective

○ **ATJX-L45-P** Allowable Displacement $F_{max}=L \times 45\%$

Model		d	Compre- ssion Length	F Max.	Allowable Load NMax.	Spring Constant N/mm	
Code	D3 L4	d	Compre- ssion Length	F Max.	Allowable Load NMax.	Spring Constant N/mm	
ATJX-L45-P	2	(5)	0.2	2.35	2.25	0.66	0.27
		(10)			4.5	1.3	
		(15)	0.26	7.8	6.7	2.0	0.28
		(20)			9.0	2.7	
		(25)	0.29	13.6	11.2	3.3	0.27
		(30)			13.5	4.0	
	3	(5)	0.26	1.8	2.25	1.1	0.47
		(10)	0.32	3.5	4.5	2.2	
		(15)			6.7	3.2	0.46
		(20)	0.35	6.3	9	4.4	
		(25)			11.2	5.5	0.47
		(30)	13.5	6.6			
	4	(35)	0.4	12.4	15.7	7.6	0.46
		(40)	18	8.8			
		(5)	0.32	2	2.25	1.1	0.47
		(10)	0.35	3	4.5	2.2	
		(15)	0.4	5.2	6.7	3.2	0.46
		(20)			9	4.4	
	5	(25)	0.45	9.5	11.2	5.5	0.47
		(30)			13.5	6.6	
(35)		0.5	15.5	15.7	7.6	0.46	
(40)				18	8.8		
(45)		0.5	15.5	20	9.8	0.47	
(50)				22.5	10.8		
6	(60)	0.55	22.6	27	12.7	0.45	
	(5)	0.35	2.01	2.25	1.1	0.47	
	(10)	0.4	3.2	4.5	2.2		
	(15)			6.75	3.3		
	(20)	0.45	4.95	9	4.4		
	(25)			11.25	5.5		
	7	(30)	0.5	7.75	13.5	6.7	0.48
		(35)			15.75	7.7	
		(40)	0.6	19.5	18	8.8	0.47
		(45)			20.25	10	
		(50)	0.65	27.95	22.5	11.2	0.48
		(60)			27	13.4	
	8	(5)	0.4	2.2	2.25	1.1	0.47
		(10)	0.5	5	4.5	2.2	
(15)		6.7			3.2		
(20)		0.55	7.7	9	4.4	0.46	
(25)				11.2	5.5		
(30)		0.6	10.8	13.5	6.6	0.47	
(35)				15.7	7.6		
(40)		0.65	15.6	18	8.8	0.47	
(45)				20	9.8		
(50)		0.7	23.1	22.5	10.8	0.46	
(60)				27	12.7		
(70)		0.75	30.8	31.5	14.7	0.45	
9		(10)	0.6	5.4	4.5	2.2	0.47
		(15)	0.65	7.2	6.7	3.2	
	(20)	0.7	10.5	9	4.4	0.47	
	(25)			11.2	5.5		
	(30)	0.75	13.5	13.5	6.6	0.46	
	(35)			15.7	7.6		
10	(40)	0.8	18.4	18	8.8	0.47	
	(45)			20	9.8		
	(50)	22.5	10.8	0.46			

Model			d	Compre- sion Length	F Max.	Allowable Load NMax.	Spring Constant N/mm
Code	D	L					
ATJX-L45-P	8	60	0.85	26.4	27	12.7	0.45
		70			31.5	14.7	
		80			36	17.7	
		10	0.65	4.6	4.5	2.2	0.47
		15			6.7	3.2	0.46
		20			9	4.4	0.47
		25	0.8	9.6	11.2	5.5	
		30			13.5	6.6	
		35			15.7	7.6	0.46
		40	0.85	12.8	18	8.8	0.47
		45			20	9.8	
		50			22.5	10.8	
		60	1.0	28	27	12.7	0.45
		70			31.5	14.7	
		80			36	17.7	
		15	0.8	6.8	6.7	3.2	0.46
		20			9.0	4.4	0.47
		25			11.2	5.5	
		30	0.9	10.8	13.5	6.6	
		35			15.7	7.6	
		40			18	8.8	0.47
		45	1.0	17	20	9.8	0.46
		50			22.5	10.8	
		60			27	12.7	
		70	1.1	26.4	31.5	14.7	0.47
		80			36	17.7	
		15			0.8	5.6	
	20	9	4.4	0.47			
	25	11.2	5.5				
		30	1.0		14	13.5	6.6
		35		15.7		7.6	
		40		18		8.8	0.47
	45	1.1	21.5	20	9.8	0.46	
	50			22.5	10.8		
	60			27	12.7		0.45
	70	1.2	30	31.5	14.7	0.47	
	80			36	17.7		
	15			0.9	5.6		6.7
20	1.0	7.7	9			4.4	0.47
25			11.2			5.5	
			30	1.1	12.1	13.5	
	35	15.7	7.6				
	40	18	8.8			0.47	
	45	1.2	18	20	9.8	0.46	
	50			22.5	10.8		
	60			27	12.7		0.45
	70	1.3	26	31.5	14.7	0.47	
	80			36	17.7		
	20			1.2	7.2		9
25	1.3	9.1	11.3			11.1	
30			13.5			13.2	
			35	1.4	11.9	15.8	15.5
	40	18	17.7				
	45	20.3	19.9				
	50	1.6	20.8	22.5	22.1	0.47	
	60			27	26.5		
	70			31.5	30.9		
	80	1.7	26.4	36	35.3		

Spring

○ **ATJX-L5-P** Allowable Displacement $F_{max}=L \times (50 \sim 60)\%$

Model		d	Compression Length	F Max.	Allowable Load NMax.	Spring Constant N/mm
Code	D ③ L ④					
ATJX-L5-P	2	5	0.18	2.0	2.5	0.49
		10			5.0	0.98
		15	0.23	6.0	7.5	1.5
		20			10	2.0
		25	0.26	11.2	12.5	2.5
		30			15	2.9
	3	5	0.23	1.6	3	0.9
		10	0.25	2.1	6	1.8
		15	0.3	4.5	9	2.6
		20			12	3.5
		25	0.32	6.4	15	4.4
		30	0.32	6.4	18	5.3
	4	5	0.26	1.4	3	0.9
		10	0.29	2	6	1.8
		15	0.32	3	9	2.6
		20			12	3.5
		25	0.38	6.1	15	4.4
		30	0.40	8	18	5.3
	5	5	0.32	1.5	3	0.9
		10	0.4	3	6	1.8
		15	0.45	7.43	15	4.4
		20			18	5.3
		25	0.5	12.25	24	7.1
		30	0.55	19.53	30	8.8
	6	5	0.32	1.5	3	0.9
		10	0.4	3	6	1.8
		15	0.5	7	15	4.4
		20			18	5.3
		25	0.55	11	24	7.1
		30	0.6	17.4	30	8.8
	8	5	0.32	1.5	3	0.9
		10	0.4	3	6	1.8
		15	0.5	7	15	4.4
		20			18	5.3
		25	0.55	9.9	24	7.1
		30	0.65	23.4	30	8.8

○ **ATJX-L45-P** Both ends are not ground for models which L specifications are marked with ()

○ **ATJX-L5-P** Both ends are ground. ○ Please use within the allowable displacement. ○ Spring Constant $N/mm=(N_{Max.})/(F_{Max.})$.

Model		d	Compression Length	F Max.	Allowable Load NMax.	Spring Constant N/mm
Code	D ③ L ④					
ATJX-L5-P	8	60	0.8	27.6	31	10.6
		70			40	12.4
		80			48	14.1
		10	0.55	3.3	6	1.8
		15	0.6	4.2	9	2.6
		20			12	3.5
	10	25	0.65	6.2	15	4.4
		30			18	5.3
		35	0.7	8.4	21	6.2
		40			24	7.1
		45	0.8	15.2	27	7.9
		50			30	8.8
	12	60	0.85	20.4	36	10.6
		70			42	12.4
		80			48	14.1
		15	0.6	3.5	9	2.6
		20	0.7	5.6	12	3.5
		25			15	4.4
	13	30	0.8	9.6	18	5.3
		35			21	6.2
		40			24	7.1
		45	0.9	12.8	27	7.9
		50			30	8.8
		60	0.9	16.2	36	10.6
	16	70	1.0	26	42	12.4
		80			48	14.1
		15	0.7	4.7	9	2.6
		20			12	3.5
		25	0.8	8	15	4.4
		30			18	5.3
	20	35	0.9	12.8	21	6.2
		40			24	7.1
		45	0.9	12.8	27	7.9
		50			30	8.8
		60	1.0	21	36	10.6
		70			42	12.4
	20	80	1.1	19	48	14.1
		15	0.75	4.3	9	2.6
		20	0.8	5.4	12	3.5
		25	0.9	7.7	15	4.4
		30			18	5.3
		35	1.0	12	21	6.2
	20	40	1.1	19	24	7.1
		45			27	7.9
		50	1.1	19	30	8.8
		60	1.2	26.4	36	10.6
		70			42	12.4
		80	1.2	26.4	48	14.1
	20	20	1.0	6	12	5.9
		25	1.1	7.7	15	7.4
		30			18	8.8
		35	1.2	10.8	21	10.3
		40			24	11.8
		45	1.3	14.3	27	13.2
	20	50			30	14.7
		60	1.4	19.6	36	17.7
		70			42	20.6
		80	1.5	27	48	23.5

○ The compression length is a reference value, and there will be a slight difference during production.

Spring